

# LOW LEVEL DESIGN (LLD)

Design & Build a RAG-Based Customer Support Assistant

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## 1. Modules

- Document Processing Module
- Chunking Module
- Embedding Module
- Vector Storage Module
- Retrieval Module
- Query Processing Module
- Graph Execution Module
- HITL Module

## 2. Data Structures

- Document -> Raw text with metadata (source, page number)
- Chunk -> Text segment (300 chars with 50-char overlap)
- Embedding -> 384-dimensional float vector
- Query -> Input string from user
- State -> Workflow data { question: str, answer: str }

### GraphState (LangGraph):

```
class GraphState(TypedDict):  
    question: str # input query  
    answer: str # output answer
```

### API Schema:

```
Input: { "query": "What is refund policy?" }  
Output: { "answer": "Refund processed within 7 days",  
          "confidence": "high" }
```

## 3. Workflow Design

LangGraph Graph Workflow:

```
START  
|  
v  
process_node (query ChromaDB, write answer to state)  
|  
|-- len(answer) >= 50 --> END (return answer)  
|
```

```
'-- len(answer) < 50 --> hitl_node --> END
```

Flow: Input -> Processing -> Conditional Route -> Output

## 4. Conditional Routing Logic

### Routing function:

```
def route_answer(state):  
    if len(state['answer']) < 50:  
        return 'hitl' # escalate  
    return 'end' # direct answer
```

### System checks:

- Confidence (answer length)
- Context availability

### Result:

- Good answer (len >= 50) -> Return to user
- Poor answer (len < 50) -> Escalate to human

## 5. HITL Design

### Triggered when:

- Answer length is less than 50 characters
- No relevant chunks found in vector store

### Process:

- Log ticket with timestamp and original query
- Notify human support agent
- Return acknowledgement message to user
- Human responds and answer is recorded

## 6. API Design

### Input:

```
{ "query": "What is refund policy?" }
```

### Output:

```
{ "answer": "Refund processed within 7 days",  
  "confidence": "high" }
```

## 7. Error Handling

- Missing PDF -> Print error message and exit
- No results found -> Return fallback response to user
- System failure -> Retry once, then log and exit gracefully

- LLM timeout -> Retry with exponential backoff (3 attempts)